

1st- vs 2nd-order a' : reduced-state estimator

0. Definitions

$$h[k] = h_d[k] - \delta x[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_x[k] := \left. \frac{da_x}{dh} \right|_{h_d[k]} = -\frac{a_x}{c} \frac{K_h}{R}, \quad K_h = \frac{1}{c} \frac{dc}{dh}$$

$$\delta x[k+1] = \lambda_c \delta x[k] - F_{dx}[k] e_{a_x}[k] - \varepsilon[k], \quad F_{dx}[k] = f_{dx}[k] + (1-\lambda_c) \sum_{i=1}^d f_{dx}[k-i]$$

$$\mathbf{x} = [\delta x_1, \delta x_2, \delta x_3, a_x, a', (\delta a')]^\top \text{ (6-state 2nd-order; 5-state 1st drops } \delta a').$$

1. True

$$x[k+1] = x[k] + a_x[k] \{f_{dx}[k] + f_T[k]\} + \delta x_D[k], \quad \delta x_D \equiv 0$$

Only random driver: thermal f_T (sensor n_x separate); trajectory h_d and curve $c(h)$ are deterministic.
 $h = h_d - \delta x$, $x_{ram} := -\delta x$, $\sigma_T^2 := \text{Var}(a_x f_T) = 4k_B T a_x$.

$$a_x = a_x(h_d) + a' x_{ram}, \quad a' = a'(h_d) + a'' x_{ram} \quad (\text{curve lifts the thermal } x_{ram})$$

Each increment = deterministic sweep ($\propto \Delta h_d$, feedforward) + thermal ($\propto \Delta x_{ram}$, the only Q):

$\begin{aligned} a_x[k+1] &= a_x[k] + a' \Delta h_d & + a' \Delta x_{ram} \\ a'[k+1] &= a'[k] + a'' \Delta h_d & + w_1, \quad w_1 := a'' \Delta x_{ram} \end{aligned}$
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a' process model on the thermal part (sweep $a'' \Delta h_d$ is feedforward):

$$\text{1st-order: } a'[k+1] = a'[k] + w_1; \quad \text{2nd-order: } a'[k+1] = a'[k] + \delta a', \quad \delta a'[k+1] = \beta \delta a' + w_2$$

$$w_1 := a'' \Delta x_{ram}, \quad w_2 := a'' \Delta^2 x_{ram}$$

2. Measurement

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + r_2[k] \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \Delta a_x[k-i] \end{bmatrix}, \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

$a', \delta a'$ have no H column: reached only through $\mathbf{F}_e(4, 5) = \Delta h_d$.

3. Esti

$$\begin{aligned} \delta \hat{x}_1[k+1] &= \delta \hat{x}_2[k], & \delta \hat{x}_2[k+1] &= \delta \hat{x}_3[k], & \delta \hat{x}_3[k+1] &= \lambda_c \delta \hat{x}_3[k] \\ \hat{a}_x[k+1] &= \hat{a}_x[k] + \hat{a}'[k] \Delta h_d[k] + (1-\lambda_c) \hat{a}'[k] \delta \hat{x}_3[k] \\ \hat{a}'[k+1] &= \hat{a}'[k] + \delta \hat{a}'[k] \quad (1\text{st: } \hat{a}'[k+1] = \hat{a}'[k]), & \delta \hat{a}'[k+1] &= \beta \delta \hat{a}'[k] \end{aligned}$$

$$\hat{\mathbf{x}}[k] = \hat{\mathbf{x}}[k|k-1] + \mathbf{L}[k](\mathbf{y}[k] - \mathbf{H} \hat{\mathbf{x}}[k|k-1])$$

4. Error

$$e_{a_x} = a_x - \hat{a}_x, \quad e_{a'} = a' - \hat{a}', \quad e_{\delta a'} = \delta a' - \delta \hat{a}', \quad e_{\delta x_i} = \delta x_i - \delta \hat{x}_i$$

$$\begin{aligned} e_{\delta x_3}[k+1] &= \lambda_c e_{\delta x_3}[k] - F_{dx}[k] e_{a_x}[k] - \varepsilon[k] \\ e_{a_x}[k+1] &= (1-\lambda_c) \hat{a}'[k] e_{\delta x_3}[k] + (1 + \hat{a}'[k] F_{dx}[k]) e_{a_x}[k] + \Delta h_d[k] e_{a'}[k] + \hat{a}'[k] \varepsilon[k] \\ e_{a'}[k+1] &= e_{a'}[k] + e_{\delta a'}[k] \quad (1\text{st: } e_{a'}[k] + w_1[k]), & e_{\delta a'}[k+1] &= \beta e_{\delta a'}[k] + w_2[k] \end{aligned}$$

5. $\mathbf{F}_e$

$$\mathbf{e} = [e_{\delta x_1}, e_{\delta x_2}, e_{\delta x_3}, e_{a_x}, e_{a'}, e_{\delta a'}]^\top$$

$$\mathbf{F}_e^{(1)} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx} & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}' & 1+\hat{a}'F_{dx} & \Delta h_d \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{F}_e^{(2)} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx} & 0 & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}' & 1+\hat{a}'F_{dx} & \Delta h_d & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & \beta \end{bmatrix}$$

$$\mathbf{q}[k] = [0, 0, -\varepsilon, \hat{a}' \varepsilon, (w_1 \text{ or } 0), w_2]^\top, \quad \mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

6. Q (thermal-grounded)

All Q traces to the thermal seed $\sigma_T^2 = 4k_B T a_x$; the curve derivatives lift the closed-loop position increments $\Delta x_{ram}, \Delta^2 x_{ram}$ to each state. Deterministic sweep ($a' \Delta h_d, a'' \Delta h_d$) is feedforward, *not* Q .

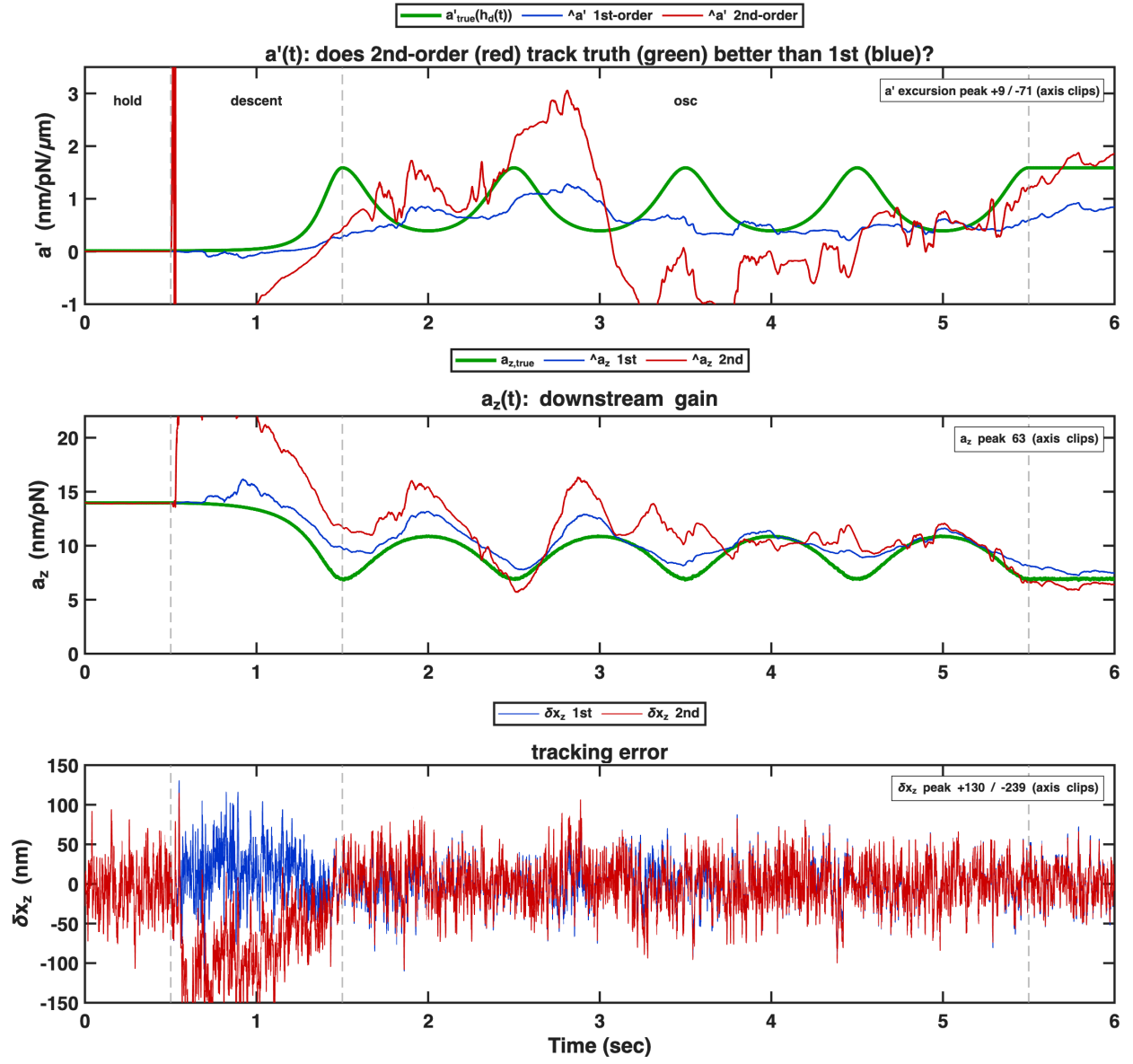
$$\text{Var}(\Delta x_{ram}) = \frac{2}{1+\lambda_c} \sigma_T^2, \quad \text{Var}(\Delta^2 x_{ram}) = (6 - 8\rho_1 + 2\rho_2) C_{\delta x} \sigma_T^2, \quad C_{\delta x} = 2 + \frac{1}{1-\lambda_c^2}$$

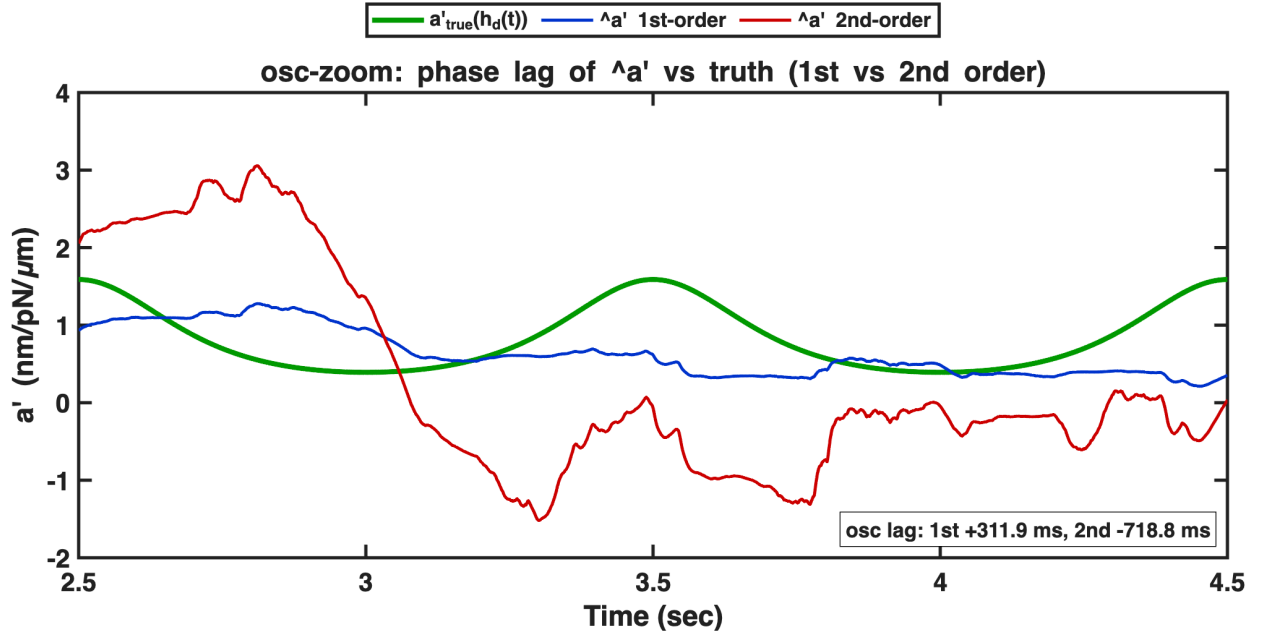
$$\begin{aligned} Q_{33} &= \sigma_T^2 & (\text{slot 3, position, direct thermal}) \\ Q_{44} &= a'^2 \text{Var}(\Delta x_{ram}) = \frac{2}{1+\lambda_c} a'^2 \sigma_T^2 & (\text{slot 4, gain via } a') \\ Q_{55} &= a''^2 \text{Var}(\Delta x_{ram}) = \frac{2}{1+\lambda_c} a''^2 \sigma_T^2 & (\text{slot 5, 1st-order, via } a'') \\ Q_{66} &= a''^2 \text{Var}(\Delta^2 x_{ram}) & (\text{slot 6, 2nd-order; } Q_{55} = 0) \end{aligned}$$

oracle (known c): $a' = -\frac{a_x}{c} \frac{K_h}{R}$, a'' from $d^2 c / d\bar{h}^2$; evaluate at $h_d[k]$. $P_{55,0} = (0.005 a_{nom} / R)^2$. Feedforwarding the a' sweep $a'' \Delta h_d$ needs a'' (curvature) $\Rightarrow$ unavailable under unknown- c ; how to handle it = the simulation choice.

7. Simulation

(Below: prior run with sweep-based oracle Q ; the thermal-grounded Q simulation of §6 is pending the sweep-handling choice.)





Canonical 1 Hz, weak-obs as-state, seed 1, small $P_{55,0}$. osc a' RMS: 2nd-order 1.17 vs 1st-order 0.55; 1st-order tracks (lag +312 ms), 2nd-order (IRW $\beta=1$) wanders -1.5 to $+3$ (free integrator, $\delta a'$ unobservable). $\hat{a}_z$ onset peak 63, a' excursion $+9/-71$ (small $P_{55,0}$).